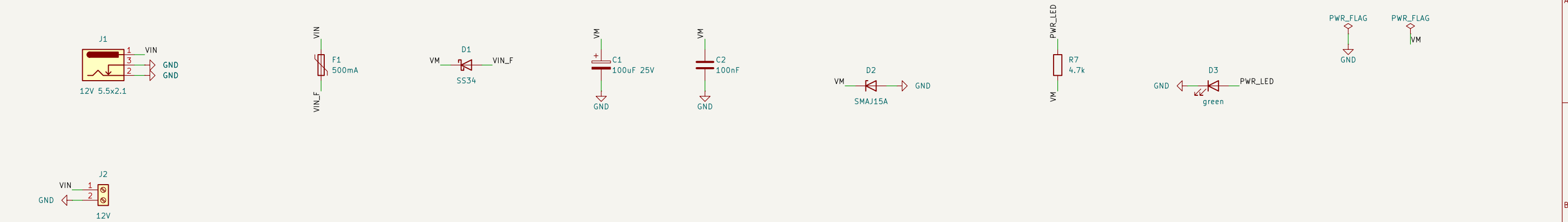


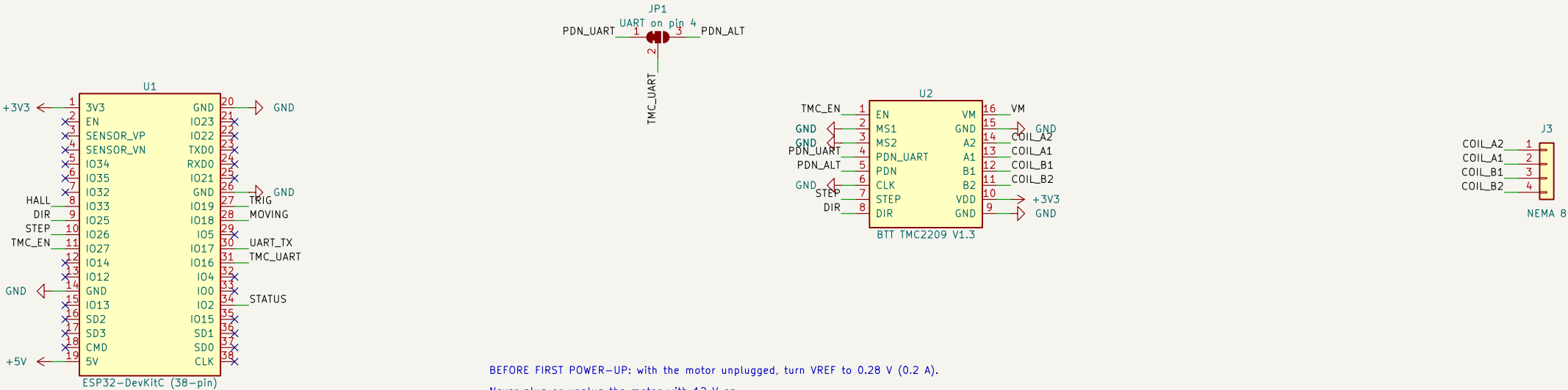
Scan-mirror controller: ESP32-DevKitC + BTT TMC2209 for the NEMA 8 mirror stepper

12 V in on J1 or J2 (ALITOVE 12 V 3 A). F1 limits it, D1 blocks reverse polarity, D2 clamps spikes under the TMC2209's 29 V limit. D3 is lit while the motor has power.

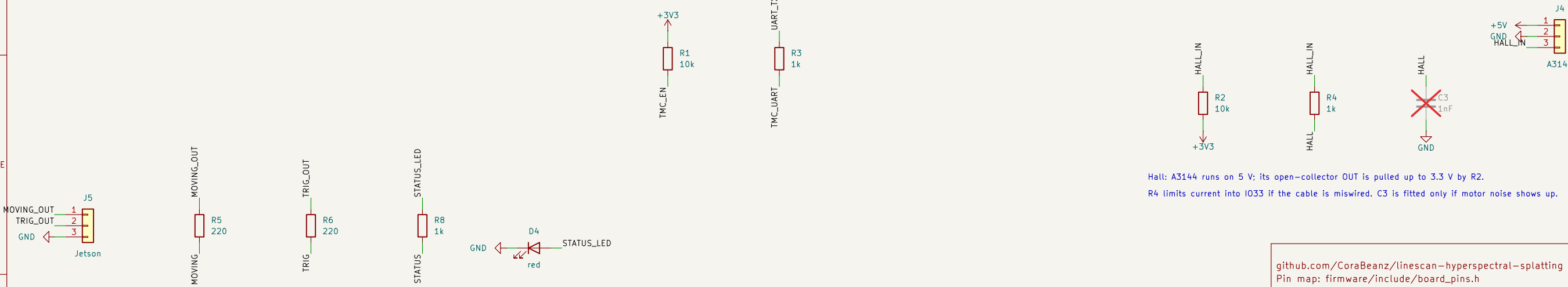


The ESP32 is powered over USB from the Jetson, which is also its command link, so there is no 5 V regulator here. With USB unplugged the TMC2209 has no logic supply (VDD), so the motor cannot move.

R1 holds EN high (motor off) while the ESP32 boots. UART: RX (IO16) straight to PDN_UART, TX (IO17) through R3. JP1 picks the module pin: 1-2 = pin 4 (BTT default), 2-3 = pin 5. MS1 = MS2 = GND: UART address 0. CLK = GND: internal clock.



BEFORE FIRST POWER-UP: with the motor unplugged, turn VREF to 0.28 V (0.2 A). Never plug or unplug the motor with 12 V on.



J5 to the Jetson's 40-pin header (3.3 V GPIO): MOVING, TRIG, GND. R5 and R6 protect both ends from a wrong pin. D4 is the firmware's status LED on IO2 (off: disabled, on: enabled, blinking: driver fault); a genuine DevKitC has no LED there. The LED and R8 keep IO2 low at boot, as it must be.

github.com/CoraBeanz/linescan-hyperspectral-splattng
Pin map: firmware/include/board_pins.h
ESP32-DevKitC + BTT TMC2209 for the scan mirror's NEMA 8
CoraBeanz
Sheet: /
File: scan_controller.kicad_sch
Title: Scan-mirror controller
Size: A3 Date: 2026-10-08 Rev: A
KiCad E.D.A. 10.0.6 Id: 1/1